Mridul Sahu

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Senior Software Engineer on Google's Core ML Frameworks team and functional lead of **Orbax** India. I direct the checkpointing function end to end: roadmap and OKR planning, work assignment and delegated design for the India team, delivery timelines for US counterparts and Google Cloud customers, and escalation ownership, on **Orbax**, JAX's distributed checkpointing library. I work the **JAX / XLA** training stack at the sharding, collective, and compiler level: checkpoint I/O, fault tolerance, and large-model save/load. Contributor to the **Orbax paper** (arXiv, 2026). 8+ years building high-performance distributed systems across ML infrastructure, planet-scale backend services, and low-latency trading platforms.

WORK EXPERIENCE

Google **Oct. 2021 – Present**

Senior Software Engineer · Orbax India Lead, Core ML Frameworks (Dec. 2024 – Present) *Bangalore, India*

- **Lead the Orbax India function, with direct ownership of its charter and planning**: run OKR planning and work assignment for the India engineering team, set delivery timelines for US counterparts and external Cloud customers, and review the team's work from design through launch.
- Set the long-term plan for **continuous checkpointing** across Google's flagship training and post-training stacks in service of company-wide TPU-efficiency goals, delegating feature design, implementation, and launch to team engineers and owning final review.
- **Direct the resolution of high-stakes Orbax escalations** for big-tech Google Cloud customers: triage and delegate fixes (slow checkpoint loads, storage optimizations, sharding bugs) to team engineers and own review and verification, unblocking training runs of models with **billions to trillions of parameters** on fleets of 32k+ TPU chips.
- Founding member of the **Core ML India** team: drove the transfer of Orbax workstream ownership from the US to India, proposed the charter of projects the team owns beyond Orbax, defined hiring signals for early hires, and mentor 8+ engineers across the Keras, JAX, and Orbax ecosystems.
- **Cut Orbax checkpoint blocking-save time 30–50%**, with substantial annualized TPU/GPU savings, by designing an asynchronous directory-creation mechanism on JAX distributed-client coordination; awarded Google's internal Perfy performance award. Now direct team engineers on follow-on save-path optimizations.
- Designed a **sharding-driven checkpoint loader** that reads from the target `jax.sharding` rather than file layout, eliminating per-tensor reshard collectives and XLA recompilation and cutting cluster read amplification to ~1× the checkpoint; direct team engineers on its follow-on sharding fixes.
- Cut large-model restart time and **storage read amplification N×→1×** with Single-Replica Restore + Broadcast (one replica loads, then broadcasts); hardened fault tolerance via the Emergency Checkpoint Manager's two-stage local + persistent save, delegating its follow-on metadata design to team engineers.
- Architected Orbax's next-generation **Dispatcher API**, unifying multi-controller and single-controller checkpointing behind one extensible interface that lets single-controller frameworks like Pathways checkpoint at large distributed scale; directed the follow-on design and development, delegated to team engineers.
- Built Orbax's checkpoint **benchmarking and observability framework** (stage timings, XProf traces, per-device HBM usage, TensorStore I/O) with A/B regression baselines, correctness digests, and CI/CD, validated at llama3-8B/70B/405B scale; delegated the design of the internal and open-source testing harnesses and Cloud benchmarking setups built on it, and administer the Cloud benchmark project, its usage policy, and request approvals.
- Built v0 of the **Knowledge Distillation** module for Tunix (Google's JAX-native LLM post-training library): a strategy-pattern framework for logit, feature/attention transfer, and pooling/projection alignment across heterogeneous architectures; directed the module's continued development by team engineers.

Software Engineer III → Senior Software Engineer, Google One (Oct. 2021 – Nov. 2024) *Bangalore, India*

Storage Quota Service · Apr. 2023 – Nov. 2024

- Designed an asynchronous processing system that eliminated database lock contention and sustained **O(100K) write QPS**, unlocking usage-calculation capacity across all Google services.
- Tech-led a Gmail × Storage Quota Service initiative for a **billion-plus-user** base: built a scalable experimentation framework, piloted a 30-day storage grace period in Gmail's codebase, and shipped email + in-Gmail warning banners alerting millions of users.
- Shipped a phased out-of-storage launch reaching millions of users each year with **zero production incidents**, driving substantial Google One signups and upgrades.
- Built a data-validation and recovery pipeline safeguarding storage-usage integrity for **billions of user accounts**, plus a precision outage-recovery mechanism restoring usage across Gmail, Photos, and Drive.

- Built real-time **service-level usage-breakdown** infrastructure for Enterprise Customers, eliminating inefficient caching of millions of user-level rows in admin consoles.

Benefits · Oct. 2021 – Apr. 2023

- Built the foundational benefits-eligibility backend (cornerstone of a major customer-acquisition feature), improving performance **70% over v1**, plus a reconciliation framework preventing data inconsistencies across partner teams (Photos, YouTube, Workspace, GStore).
- Led multiple benefit launches and international expansions impacting **millions of users** as central technical liaison across Product Areas; built backend to dynamically split/combine benefits for tailored delivery.
- Fixed critical infrastructure deficiencies affecting millions of users; established alerting/SLO/monitoring standards; owned a **zero-incident on-call migration** of the Benefits and Storage Quota Service teams from the US to India.

Zopte Technologies (CrispyGames) **2018 – 2021**
Software Engineer *Mumbai, India*

- Engineered a Python backtesting and live-trading platform (Bloomberg, Zerodha, NSE-NNF, QuickFIX, custom TimeSeries DB) with an **~80× performance uplift**, plus a distributed backtesting optimizer for strategy-parameter search over a compute cluster.
- Built a low-level **NSE/NFO exchange** library (proprietary protocols, binary structures, security compliance), an Order Management System (OMS), and Tick Collector; maintained an NSE-licensed CTCL (computer-to-computer link) trading application in production at a brokerage.
- Built a code generator producing customizable, mockable server- and client-side codebases (up to **60% of project code**), wired to a custom real-time continuous-deployment system.
- Architected a **microservices service mesh** from scratch on Kubernetes with GCP and Istio (gRPC/REST, Zookeeper, Redis, Elasticsearch, MongoDB); built pluggable Flutter / Angular Dart UI components.

PROJECTS

research.rudrite.com · Interactive Research-Paper Visualizer **2025 – Present**

- Building a site that **visualizes and explains landmark AI/ML research papers** through 120+ interactive, animated explainers spanning training-at-scale parallelism (Megatron, GSPMD, ZeRO), efficient LLM serving (batching, speculation, KV-cache), transformers, and RLHF / reasoning.

explainers.rudrite.com · Interactive Visual Explainers **2026 – Present**

- Building a library of **30+ interactive visual explainers** of system design, distributed systems, and real code, from high-level architecture to line-by-line code walks.

kernels.rudrite.com · TPU Kernel & ML-Systems Mastery Courses **2026 – Present**

- Building four open, lab-driven mastery courses on the TPU ML stack (**Pallas kernels, JAX, XLA, PyTorch**): 53 chapters, 30 runnable labs, drills, failure exhibits, and live instruments, every number benchmarked on real chips.

PUBLICATIONS

- C. Gaffney, S. Li, D. Ng, **M. Sahu**, et al., **"Orbax: Distributed Checkpointing with JAX,"** *arXiv:2605.23066* (cs.DC / cs.LG), 2026. A modular, JAX-native checkpointing library demonstrating up to **3.5x faster saving and 2x faster loading** vs. comparable PyTorch checkpointing.
- S. K. Jaiswal, M. Pal, **M. Sahu**, P. Sahu, A. Dev, "EvoCut: A New Generalization of the Albert-Barabási Model for Evolution of Complex Networks," *22nd FRUCT Conference*, Jyväskylä, 2018, pp. 67–72.

EDUCATION

National Institute of Technology, Meghalaya **2014 – 2018**
B.Tech, Computer Science & Engineering, CGPA 8.76/10 *Shillong, India*

TECHNICAL SKILLS

Languages: Python, Java, Go, C++, Dart, JavaScript

ML & Performance: JAX, XLA, Orbax, Flax (NNX), Optax, Tunix, TensorStore, Pathways on Cloud; distributed training, sharding, collectives, TPU, profiling (XProf, jax.profiler, jax.monitoring), knowledge distillation, ML systems

Technical Leadership: roadmap & OKR planning, work assignment & delegated design, design/launch review, escalation direction, cross-org & customer timeline ownership, hiring signals, mentoring

Systems & Infrastructure: distributed systems, microservices, system design, API design, low-latency / trading systems (FIX, OMS, exchange protocols), gRPC, REST; Kubernetes, GCP, Docker, XPK

Data & Web: SQL, MySQL, MongoDB, Elasticsearch, Redis, Cloud Pub/Sub; Flutter, Angular Dart, OpenCV